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Web Technology

By Dr. Ashutosh Kumar Bhatt | Uttarakhand Open University, Haldwani

Learners enrolled: 10843


Web Technology refers to the many different tools and techniques that are utilized in the process of communication between different types of devices over the internet in Hindi/English Language. The main sections or components of Web Technology are World Wide Web (WWW), Web Browser, Web Server, Web Pages, and Web Development. It can be classified into two ways: Frontend and Backend Development. The part of a website where the user interacts directly is termed as front end. It is also referred to as the client-side of the application. Backend is the server side of a website. It is the part of the website that users cannot see and interact with. It is the part of software that does not come in direct contact with the users. It is used to arrange/store data. The frontend languages are HTML, CSS, JavaScript, AJAX, etc. and Backend Languages are PHP, Node.js, Python, Ruby, Java, JavaScript, C# etc.

Summary

Course Status :	Completed
Course Type :	Not Applicable
Language for course content :	English
Duration :	12 weeks
Category :	Computer Science and Engineering
Level :	Undergraduate/Postgraduate
Start Date :	01 Jan 2024
End Date :	30 Apr 2024
Enrollment Ends :	29 Feb 2024
Exam Date :	19 May 2024
	IST
Shift :	Shift-I

Note: This exam date is subject to change based on seat availability. You can check final exam date on your hall ticket.

Course layout

Week	Topic	Module
Week 1	MARKUP LANGUAGES Web Standards, The History of Markup Languages HTML, XML, XHTML	MARKUP LANGUAGES Web Standards, The History of Markup Languages
	WML-Wireless Markup Language Cascading Style Sheets (CSS) DHTML	HTML, XML, XHTML
	Client-Side and Server-Side Technologies Client-Side and Server-Side Technologies Basics of HTML	WML-Wireless Markup Language, Cascading Style Sheets (CSS)
	HTML Document Structure	DHTML and Client-Side and Server-Side Technologies

		Basics of HTML
		HTML Document Structure
Week 2	HTML, Introduction to Hyperlinks, Hyperlinks - Examples Links to the same document, HTML, Frames and Framesets Frame attributes Nested framesets, HTML Tables, Tables Attributes , Rows and Columns	HTML
		Introduction to Hyperlinks, Hyperlinks - Examples Links to the same document
		HTML, Frames
		Framesets
		Frame attributes Nested framesets
		HTML Tables, Tables Attributes , Rows and Columns
Week 3	Introduction to Forms The <FORM>, <INPUT> Tag Text input, CASCADING STYLE SHEETS (CSS) Inline Style sheet , Internal Style Sheet , External Style sheet, CASCADING STYLE SHEETS, Imported Style Sheet ,Cascading Rules , Order rules , Order rules	Introduction to Forms
		The <FORM>, <INPUT> Tag Text input
		CASCADING STYLE SHEETS (CSS) Inline Style sheet ,
		Internal Style Sheet , External Style sheet
		CASCADING STYLE SHEETS, Imported Style Sheet
		Cascading Rules , Order rules , Order rules
Week 4	CSS, Difference between ID and Classes , Anatomy of a style rule , Style Classes , Attribute Selector XHTML – eXtensible Hyper Text Markup Language , Need for XHTML XHTML – Base syntactic rules , XHTML – other syntactic rules , DTD – Document Type Definition	CSS, Difference between ID and Classes
		Anatomy of a style rule , Style Classes , Attribute Selector
		XHTML – eXtensible Hyper Text Markup Language
		Need for XHTML
		XHTML – Base syntactic rules , XHTML – other syntactic rules
		DTD – Document Type Definition
Week 5	XML ,XML Trees Namespaces , Defining XML Data Formats JAVASCRIPT AND HTML Document Object Model or DOM	XML
		XML Trees
		Namespaces
		Defining XML Data Formats
		JAVASCRIPT
		HTML
		Document Object Model or DOM
Week 6	JavaScript - Document Object Model or DOM , Types of DOM nodes JavaScript - Traversing the DOM tree , Working with DOM JAVASCRIPT AND EVENT HANDLING- JavaScript - Event Handlers	JavaScript - Document Object Model or DOM
		Types of DOM nodes
		JavaScript - Traversing the DOM tree
		Working with DOM
		JAVASCRIPT AND
		EVENT HANDLING
		JavaScript - Event Handlers
Week 7	JAVASCRIPT AND EVENT HANDLING- onClick event type , onSubmit event type	JAVASCRIPT AND EVENT HANDLING

	JAVASCRIPT AND EVENT HANDLING- onMouseOver and onMouseOut event JAVASCRIPT AND EVENT HANDLING- java script cookies , create cookies	onClick event type
		onSubmit event type
		JAVASCRIPT AND EVENT HANDLING
		onMouseOver and onMouseOut event
		JAVASCRIPT AND EVENT HANDLING
		java script cookies , create cookies
Week 8	JAVASCRIPT AND EVENT HANDLING- javascript cookies , Store Cookies JAVA SERVER PAGES(JSP), Introduction to JSP , Comparison of Servlets with JSP Advantages of JSP , JSP engines , JSP Architecture	JAVASCRIPT AND EVENT HANDLING- javascript cookies
		Store Cookies
		JAVA SERVER PAGES(JSP)
		Introduction to JSP
		Comparison of Servlets with JSP
		Advantages of JSP
		JSP engines, JSP Architecture
Week 9	Life cycle of a JSP Page , Directory structure of JSP , Anatomy of a JSP page JSP Components and Tags JSP Creation of a small Web application using JSP	Life cycle of a JSP Page
		Directory structure of JSP
		Anatomy of a JSP page
		JSP Components
		JSP Tags
		JSP Creation
		A Small Web application using JSP
Week 10	JAVA SERVER PAGES(JSP) , JSP Elements – JSP Actions JSP Implicit Objects JSP - Exception Handling	JAVA SERVER PAGES(JSP) Basics
		JSP Elements –
		JSP Elements Examples
		JSP Actions
		JSP Actions Examples
		JSP Implicit Objects
		JSP - Exception Handling
Week 11	JSP Session Tracking JAVA SERVER PAGES (JSP) WITH JDBC- Basic steps to connect database , Install Mysql , Steps to run JSP using Tomcat server Creating a table using JSP example , Creating a table: database	JSP Session Tracking
		JAVA SERVER PAGES (JSP) WITH JDBC
		Install Mysql
		Steps to run JSP using Tomcat server
		Basic steps to connect database
		Creating a table using JSP

		example
		Creating a table: database
Week 12	JAVA SERVER PAGES (JSP) WITH JDBC- SELECT operation , INSERT operation , DELETE Operation JDBC – INTRODUCTION , JDBC , JDBC Architecture JDBC Drivers , Working with JDBC, JDBC - JDBC Object Classes JAVA SERVLET IMPLEMENTATION, Tomcat Installation JAVA SERVLET -Servlet Implementation JAVA SERVLET -The doGet method , The doPost method	JAVA SERVER PAGES (JSP) WITH JDBC- SELECT operation , INSERT operation , DELETE Operation
		JDBC – INTRODUCTION , JDBC , JDBC Architecture
		JDBC Drivers , Working with JDBC
		JDBC - JDBC Object Classes
		JAVA SERVLET IMPLEMENTATION, Tomcat Installation
		JAVA SERVLET -Servlet Implementation
		JAVA SERVLET -The doGet method , The doPost method

Books and references

- Powell. Thomas A., JavaScript: The Complete Reference
- Iemay. Laura, rafe colburn, jennifer kyrnin, Mastering HTML, CSS & JavaScript Web, BPB Publication, 2016
- Vishvajeet. Sisodia, Basic of Web Design, HTML, CSS3, Centrum Press, 2014.

Instructor bio



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Dr. Ashutosh Kumar Bhatt is working in School of Computer Science and Information Technology, Uttarakhand Open University, Haldwani, Distt- Nainital (Uttarakhand). Dr. Bhatt has completed Ph. D. in Computer Science in 2009. His work area of research was Artificial Neural Network.

He has more than Seventeen years of teaching and research experience in various organizations of repute for PG & UG courses of Computer Science & IT. He is also associated with many renowned National/ International Journals as Lead Guest Editor/reviewer/editorial board member. More than 12 SCIE/ESCI and more than 55 Scopus indexed research publications are credited to him in reputed National/International Journals. He is a life member of CSI(Computer Society of India) and member of IEEE. He had served as State Student Coordinator of Region I Uttarakhand of CSI (for three years) and also served as Secretary of Uttarakhand ACM Professional Chapter. He is awarded by the significant contribution award by CSI for the year 2012-13 and Distinguished Professor award by CSI Mumbai Chapter. He has published and edited number of books e.g. C# Programming using Dot Net Framework, Technology Enabled Learning, Introduction to Computers with basics of Programming, Artificial Intelligence for Societal Development and Global Well-Being etc. He had worked in a project entitled “Automated Analysis for Quality Assessment of Apples using Artificial Neural Network”, under DST New Delhi. He had also worked in the project entitled as “Automated IoT Based Smart Water Quality Assessment System” “Medicinal Knowledge Portal Exclusively for Uttarakhand